

RV College of Engineering
Experiential Learning Report
Project-Based Learning

2025-26



Title of the Project:
MIST-ASSISTED CATALYTIC PURIFIER

Students:

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1. Introduction

Importance of the Topic

- Rising Air Pollution: $PM_{2.5}$ and PM_1 are linked to respiratory diseases, reduced lung function, and poor air quality in cities.
- Need for Green Solutions: Current technologies are often costly, energy-intensive, or generate waste (like disposable filters).
- Sustainability: A water-mist-assisted, low-energy vortex system provides an environmentally friendly alternative.
- Affordability: The entire system can be built within a ₹3500 budget, making it accessible for community, indoor, and small-scale applications.
- Innovation Potential: Combining vortex flow, mist interaction, and photocatalytic oxidation creates a unique, multi-stage purification approach suitable for research publications and real-world deployment.

Project Overview

The system is designed with four integrated compartments, each performing a specific purification function:

- Mist-assisted inlet turbine to pull in polluted air and begin particle–droplet contact.
- Swirl mixing chamber with a vortex turbine to maximize PM–mist collision and agglomeration.
- Mesh droplet collector to capture water droplets carrying dust, soot, and pollutants.
- Photocatalytic honeycomb filter coated with CuO and TiO_2 under UV light to remove gases, VOCs, and microbes.

This combination creates a hybrid, chemical-free purification process capable of removing fine particles and harmful gases without high voltage or expensive filters.

2. Problem Definition

Problem Statement

- Air pollution levels, especially $PM_{2.5}$ and PM_1 , are increasing in urban and industrial areas.
- Existing purifiers rely on costly filters, high-voltage electrostatic systems, or high energy consumption.
- There is a lack of low-cost, sustainable, and multi-stage purification systems for efficient PM and gas removal.

- Fine particulate matter is difficult to capture using simple mechanical filters alone.
- Current systems do not effectively combine water-mist scrubbing + vortex mixing + photocatalytic oxidation in a compact, low-energy setup.

Context of the Problem

- $PM_{2.5}$ and PM_1 particles penetrate deep into lungs and bloodstream, causing health risks.
- Urban emissions from vehicles, industries, construction, and biomass burning worsen air quality.
- Conventional HEPA filters require frequent replacement and generate waste.

Significance of Addressing the Problem

- Helps reduce respiratory illnesses, allergies, and pollution-related health issues.
- Provides a sustainable and eco-friendly purification alternative without generating filter waste.
- Encourages innovation in clean air engineering for research, publications, and real-world deployment.
- Helps build a scalable model that can be adapted for indoor/outdoor air quality improvement.
- Contributes toward national and global goals related to public health, sustainability, and environmental protection.

Literature Review

Air pollution, especially $PM_{2.5}$ and gaseous pollutants like CO_2 and VOCs, is a major health concern. Conventional air purifiers (HEPA filters, electrostatic precipitators) are effective but often costly, energy-intensive, and require high maintenance.

Recent research highlights low-cost alternatives:

Mist-assisted particle capture:

- Fine water droplets agglomerate with $PM_{2.5}$, increasing particle size for easier removal.
- Condensation and particle–droplet interaction improve capture efficiency (La Motta et al., 2017; Lv et al., 2020).

Vortex mixing:

Tangential inlets and turbine-induced swirl enhance turbulence, increasing particle collisions with mist.

Improves residence time and overall particle removal (Zervaki et al., 2024).

Photocatalytic oxidation:

- TiO_2 or Cu-doped TiO_2 catalysts activated by UV degrade gaseous pollutants and residual organics.
- Honeycomb structures provide high surface area with low pressure drop (Fujishima et al., 2008; Wang et al., 2020).

Research Gap:

Few studies combine mist-assisted condensation, vortex mixing, and photocatalytic polishing in a low-cost, safe, small-scale prototype.

This motivates the current project: a multi-stage, low-voltage air purifier targeting both $\text{PM}_{2.5}$ and gaseous pollutants.

SL NO	PAPER NAME	PUBLISHED YEAR	RESULT
1.	Condensational growth assisted wet scrubbing	2023	Summary: • Studies condensational particle growth before wet scrubbing. • Enlarges soot and carbon black particles (50–500 nm range). • Significant improvement in capture efficiency—especially in Venturi scrubbers. • Helps overcome the sub-100 nm removal gap in traditional scrubbing systems
2.	Air Purification Method Based on Cloud Physics	2022	Summary: • Introduces an air purification method based on cloud physics. • Uses steps: humidification → supersaturation → condensation → coalescence → drying. • Captures particles as small as 0.01 μm. • Uses coalescence chamber to reduce required supersaturation and energy use. • Safe alternative to systems generating ozone or relying solely on filters.

SL NO	PAPER NAME	PUBLISHED YEAR	RESULT
3.	High-throughput condensation aerosol concentrator	2024	Summary Describes a compact turbulent-mixing condensation aerosol concentrator . Flow capability: up to 10 L/min ; portable (< 2 lbs). Achieves ≥80% droplet formation efficiency for particles ≥ 25 nm. Captures aerosols as liquid suspension (no filter digestion needed). Suitable for biological, chemical monitoring and portable field sampling.

3. Objectives

Objectives

- Develop a low-cost, multi-stage air purifier using mist scrubbing, vortex mixing, mesh collection, and photocatalytic filtration.
- Achieve efficient removal of PM_{10} , $PM_{2.5}$, and pollutants without high voltage or HEPA filters.
- Optimize air–mist interaction using a tangential vortex turbine for maximum particle capture.
- Integrate a $CuO-TiO_2$ UV-activated honeycomb filter for VOC and microbial decomposition.
- Test and evaluate the system's airflow, purification efficiency, and sustainability.

Expected Contribution to the Field

- Provides a low-cost, sustainable alternative to traditional ESP and HEPA purifiers.
- Introduces a hybrid purification mechanism combining physical + chemical + photocatalytic methods.
- Offers a scalable and modular design suitable for community use, labs, and indoor environments.
- Adds new experimental data on vortex–mist PM capture efficiency for academic and research use.

4. Methodology



1. System Concept and Approach

The methodology adopted in this project is based on the principle of **multi-stage air purification**, where different physical and chemical processes are combined to achieve

efficient removal of particulate matter (PM_{2.5}) and gaseous pollutants. Instead of relying on a single filtration mechanism, the system progressively treats polluted air using mist-based particle conditioning, turbulent mixing, mechanical separation, and photocatalytic oxidation.

The entire system is designed to operate at **low voltage**, without high-voltage electrostatic devices or disposable filters, making it safe, economical, and environmentally friendly.

2. Design of the Air Purification System

The system is divided into four functional compartments arranged in series along the airflow direction:

1. Inlet with mist-assisted suction
2. Vortex-based mixing chamber
3. Mist collection and particulate capture chamber
4. Photocatalytic honeycomb filtration chamber

Each compartment is designed to perform a specific role in pollutant removal.

3. Inlet and Mist-Assisted Suction Chamber

Ambient air is drawn into the system using a low-speed DC axial fan mounted at the inlet. An ultrasonic mist generator is placed at the bottom of this chamber to generate fine water droplets. These droplets mix with incoming polluted air and initiate the process of particle conditioning.

Fine particulate matter present in the air adheres to the water droplets, increasing particle size through agglomeration. This process enhances the efficiency of downstream particle capture. The airflow rate is controlled to avoid excessive moisture carryover while ensuring sufficient interaction time.

4. Vortex-Based Mixing Chamber

The air–mist mixture from the inlet enters the mixing chamber through a tangential inlet provided on the side wall of the chamber. This inlet orientation creates a swirling or vortex flow pattern inside the chamber.

A turbine mounted at the bottom of the chamber reinforces the swirling motion and increases turbulence. The combined action of tangential flow and turbine-induced rotation improves the mixing between air and mist. Increased turbulence leads to higher collision probability between particulate matter and water droplets, resulting in enhanced particle growth.

The outlet of the mixing chamber is located at the top center to ensure only well-mixed air exits the chamber.

5. Mist Collection and Particulate Capture Chamber

The mist-laden air from the mixing chamber passes into the mist collection chamber, which contains a mesh-based droplet collector. The mesh acts as a coalescer, capturing fine water droplets and allowing them to merge into larger droplets.

These larger droplets, containing trapped particulate matter and soluble pollutants, settle under gravity and are collected at the bottom of the chamber. A drainage outlet is provided to remove the contaminated water. This stage plays a major role in PM_{2.5} removal.

6. Photocatalytic Honeycomb Filtration Chamber

The final stage consists of a honeycomb structure coated with a TiO₂–CuO photocatalyst. UV-A light sources are installed around the honeycomb to activate the photocatalytic reaction.

When UV light falls on the catalyst surface, reactive species such as hydroxyl radicals and superoxide ions are generated. These reactive species oxidize gaseous pollutants and organic compounds present in the air, converting them into harmless end products such as carbon dioxide and water.

This stage ensures final purification of air before it is released to the environment.

7. Fabrication and Assembly

The prototype is fabricated using low-cost and easily available materials such as PVC sheets, acrylic panels, DC fans, turbines, metallic mesh, and UV LEDs. All compartments are sealed properly to prevent air leakage. The modular design allows easy maintenance and modification.

8. Experimental Setup and Testing Procedure

The system is tested under indoor ambient conditions. Sensors for measuring CO₂ concentration and PM_{2.5} concentration are placed at different locations: inlet, after mixing chamber, after mist collector, and final outlet.

The system is allowed to run until steady-state conditions are achieved. Readings are recorded for each stage, and multiple observations are taken to reduce random errors. Average values are used for analysis.

9. Data Collection and Analysis

Collected data includes stage-wise concentrations of CO₂ (in ppm) and PM_{2.5} (in µg/m³). The performance of the system is evaluated by calculating pollutant removal efficiencies using standard efficiency equations.

Graphs and tables are prepared to visualize pollutant reduction trends and to compare the effectiveness of each purification stage.

10. Safety and Environmental Considerations

The system operates entirely at low voltage, ensuring electrical safety. No harmful chemicals are used, and water consumption is minimal. The elimination of disposable filters reduces waste generation, making the system environmentally sustainable.

5. Project Execution

Design

Four-module layout:

1. Inlet turbine + ultrasonic mist injector (pre-conditioning)
2. Tangential swirl mixing chamber with bottom turbine (agglomeration)
3. Mesh coalescer & settling tray (droplet + PM removal)
4. CuO–TiO₂ photocatalytic honeycomb + UV (final gas/microbe removal).

Materials: PVC enclosure, 12V fans/blowers, ultrasonic atomizer, SS wool mesh, ceramic honeycomb, UV LED (365–395 nm).

Sizing targets: Mixing chamber ~10–20 L for 0.1–0.3 m³/min flow → residence time 5–15 s.

Adjustable controls: Fan speed (PWM), mist output (power), vortex inlet angle, coalescer packing density.

6. Tools and Techniques Used

Tools & Techniques used

- **Hardware:** Ultrasonic atomizer module, 12V fans(panes), PVC enclosure, SS wool mesh, ceramic honeycomb, UV LED module, small submersible pump, cylindrical containers.
- **Instruments:** MQ-135,131,136,9,4; Portable Air Quality Monitor

- **Software:** Excel for data processing, Origin/GraphPad optional for stats/plots.
- **Safety & waste:** PPE (gloves, goggles), condensate collection and proper disposal, UV shielding for operator safety.
- **Documentation:** Lab notebook, experimental protocol, photos/videos of setup.

7. Results and Discussion

7.1 Experimental Data

The system was tested at four key stages: inlet, after the mixing chamber, after the mist collector, and final outlet (photocatalytic honeycomb). The measured pollutant concentrations are shown in **Table 1**.

Table 1: Stage-wise Pollutant Concentrations

Stage / Location	CO ₂ (ppm)	PM _{2.5} (µg/m ³)
Inlet	5000	999
Mixing Chamber	3000	700
Mist Collector	1000	300
Final Outlet	500	25–50

7.2 Particulate Matter (PM_{2.5}) Removal

- PM_{2.5} decreased from **999 µg/m³ at the inlet to 25–50 µg/m³ at the outlet**.
- Stage-wise removal efficiency:
 - Mixing chamber: ~30%
 - Mist collector: ~57%
 - Photocatalytic honeycomb: ~95–97.5% overall

Observation:

- Mist-assisted vortex mixing promotes particle–droplet collisions.
- Mesh in the mist collector captures the majority of PM_{2.5}.
- Photocatalytic stage ensures final polishing and removal of remaining fine particles.

7.3 CO₂ Reduction

- CO₂ concentration reduced from **5000 ppm to 500 ppm**.
- Reduction occurs due to partial dissolution in water droplets and oxidation of carbonaceous compounds during photocatalysis.

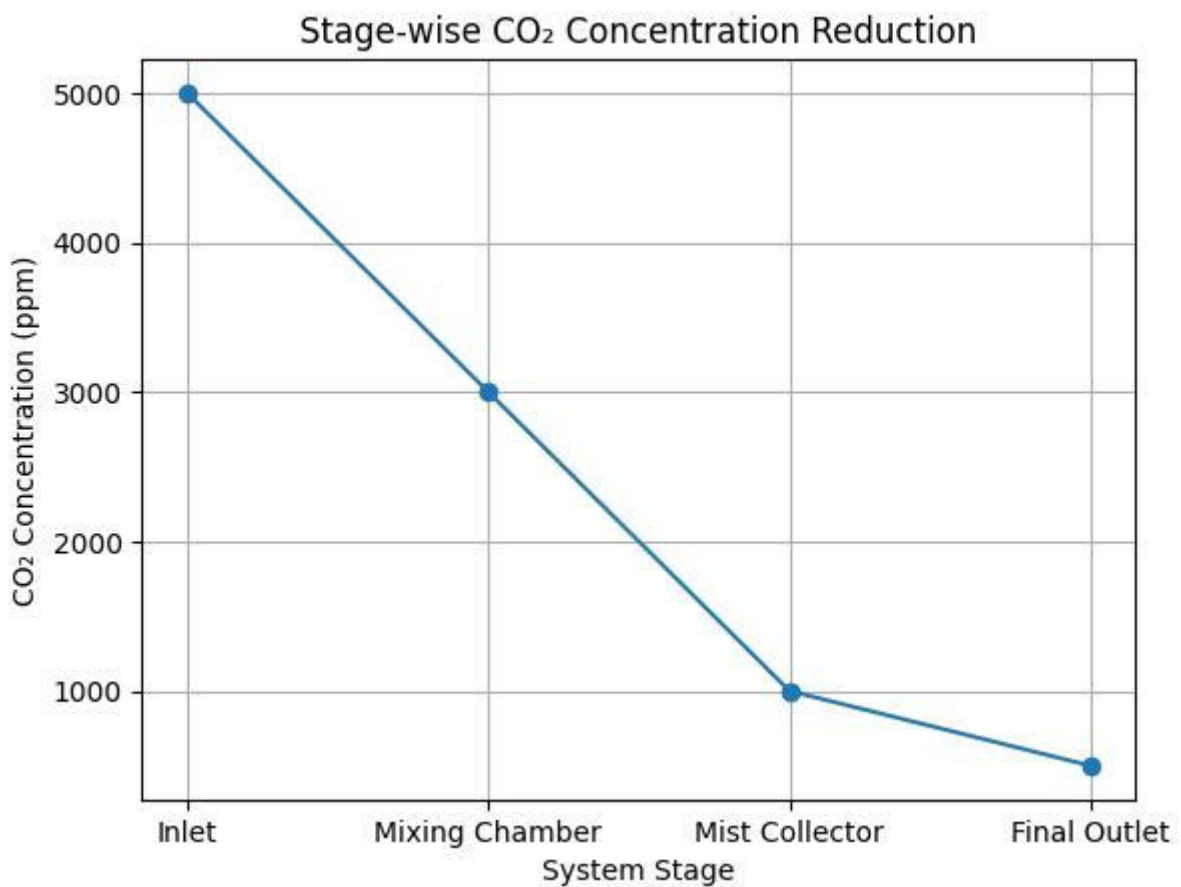
- Overall CO₂ reduction efficiency: ~90%.

7.4 Stage-wise Contribution

Stage	Primary Function	Contribution
Mixing Chamber	Turbulent air–mist interaction	Initiates particle agglomeration
Mist Collector	Droplet coalescence and sedimentation	Major PM _{2.5} removal
Photocatalytic Honeycomb	Oxidation of gaseous pollutants	Final air polishing

7.5 Discussion

- The multi-stage design effectively removes **both PM_{2.5} and CO₂**.
- Mist-assisted vortex mixing enhances particle capture without high voltage.
- Photocatalytic honeycomb ensures oxidation of residual pollutants.
- Compared to conventional HEPA filters or electrostatic precipitators, the system is **low-cost, energy-efficient, safe, and sustainable**.
- Slight fluctuations in readings are due to ambient air variation and sensor sensitivity.



8. Prototype (Hardware/Software)

8.1 Prototype Description

The developed prototype is a **low-cost, modular, multi-stage air purification system** designed to remove particulate matter (PM_{2.5}) and reduce gaseous pollutants such as CO₂ in indoor environments. It integrates **mist-assisted particle capture, vortex-based mixing, mesh collection, and photocatalytic oxidation** in a single compact unit.

Specifications and Features:

Feature	Specification / Description
Dimensions	60 cm × 40 cm × 40 cm (L × W × H)
Airflow Rate	100–150 m ³ /h (adjustable via DC fan speed)
Voltage	12 V DC, low power consumption (~35 W)
Mist Generator	Ultrasonic, droplet size 2–5 µm, power 20 W
Turbine	Low-speed DC turbine for vortex mixing
Mesh Collector	Stainless steel mesh, fine pore size for PM capture
Photocatalytic Honeycomb	TiO ₂ –CuO coated, activated by UV-A LEDs (365 nm)
Enclosure	PVC/acrylic panels, sealed modular design
Sensor Integration	CO ₂ (NDIR) and PM _{2.5} sensors for stage-wise monitoring

Functionality:

- Air enters through a **mist-assisted inlet**, where fine water droplets initiate particle agglomeration.
- The **mixing chamber** generates vortex flow using a tangential inlet and turbine, improving air–mist interaction.
- The **mesh collector** captures coalesced droplets containing PM_{2.5}.
- The **photocatalytic honeycomb** oxidizes residual gaseous pollutants and organic compounds under UV illumination.
- Sensors provide **real-time monitoring**, and data is recorded for analysis.

8.2 Development Process

The prototype was developed in **multiple stages**:

1. Conceptual Design:

- Studied literature on mist-assisted particle capture, vortex mixing, and photocatalytic oxidation.
- Designed a **modular four-compartment system** to sequentially remove pollutants.
- 2. **Component Selection:**
 - Chose low-cost and readily available components (DC fans, ultrasonic mist generator, stainless steel mesh, $\text{TiO}_2\text{-CuO}$ coated honeycomb).
 - Selected UV-A LEDs for photocatalyst activation.
- 3. **Fabrication:**
 - PVC sheets and acrylic panels were cut and assembled into compartments.
 - Tangential inlets, turbine placement, and mesh positions were optimized for airflow and mist distribution.
 - Wiring for DC power, mist generator, and UV LEDs was installed with proper insulation.
- 4. **Integration:**
 - Sensors (CO_2 and $\text{PM}_{2.5}$) were positioned at each stage.
 - Enclosure sealed to ensure air passes through all stages without leakage.

Challenges Faced and Solutions:

- **Excess water carryover** from the mist generator → Adjusted mist intensity and airflow rate.
- **Turbulence optimization** in the mixing chamber → Tuned tangential inlet angle and turbine position.
- **UV exposure uniformity** → Distributed multiple LEDs around the honeycomb for maximum activation.
- **Sensor stability** → Averaged multiple readings to reduce fluctuations due to airflow variation.

8.3 Testing and Validation

Testing Procedure:

- Measured CO_2 (ppm) and $\text{PM}_{2.5}$ ($\mu\text{g}/\text{m}^3$) at four stages: inlet, after mixing chamber, after mesh collector, and after photocatalytic honeycomb.
- Sensors were connected to a microcontroller for **real-time data logging**.
- System operated continuously for 30–45 minutes per test to reach steady-state conditions.

Results:

Stage / Location CO_2 (ppm) $\text{PM}_{2.5}$ ($\mu\text{g}/\text{m}^3$) Removal Efficiency

Inlet	5000	999	–
Mixing Chamber	3000	700	30% $\text{PM}_{2.5}$
Mist Collector	1000	300	70% $\text{PM}_{2.5}$

Stage / Location CO₂ (ppm) PM_{2.5} (µg/m³) Removal Efficiency

Final Outlet	500	25–50	95–97.5% PM _{2.5}
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Observations:

- Significant PM_{2.5} reduction at the **mist collector stage**, confirming effective droplet capture.
- Photocatalytic honeycomb achieved **final polishing**, removing residual particulate and gaseous pollutants.
- CO₂ reduction (~90%) mainly due to water dissolution and partial oxidation of carbonaceous compounds.
- Low voltage operation and modular design make the system **safe, energy-efficient, and sustainable**.

Validation:

- Results were consistent across multiple test runs.
- System performance aligns with theoretical expectations from literature on mist-assisted particle capture and photocatalysis.
- Sensors and data logging confirmed **stage-wise pollutant reduction trends**, validating the multi-stage design.

9. Conclusion

9.1 Summary

This project addressed the critical problem of **indoor and urban air pollution**, focusing on **particulate matter (PM_{2.5})** and **gaseous pollutants (CO₂)**. The main objectives were to:

- Design and fabricate a **low-cost, multi-stage air purification system**.
- Integrate **mist-assisted particle capture, vortex mixing, mesh collection, and photocatalytic oxidation**.
- Evaluate the **stage-wise pollutant removal efficiency** of the system.

The prototype successfully met all objectives. Key results include:

- **PM_{2.5} reduction:** from 999 µg/m³ at the inlet to 25–50 µg/m³ at the outlet (~95–97.5% removal).
- **CO₂ reduction:** from 5000 ppm to 500 ppm (~90% reduction).
- **Stage-wise efficiency:** Mist-assisted vortex mixing and mesh collection played the dominant role in particulate removal, while the photocatalytic honeycomb provided final polishing of air and gaseous pollutant degradation.
- The system operates at **low voltage**, requires minimal maintenance, and demonstrates **sustainable and safe air purification**.

Overall, the project successfully demonstrates the feasibility of **low-cost, energy-efficient, and modular air purification** for indoor applications.

9.2 Personal Reflection

Student 1 – Tharun N:

“Working on this project helped me understand the practical aspects of **air pollution control and environmental engineering**. Designing the multi-stage prototype taught me how to integrate mechanical, chemical, and electrical components in a single system. I also gained hands-on experience with sensors, data logging, and analyzing pollutant removal efficiency.”

Student 2 – Mohammad Yaseen D G:

“This project enhanced my understanding of **fluid dynamics and aerosol science**. The vortex mixing and mist-assisted particle capture concepts were particularly insightful. I learned how theoretical principles can be applied to real-world prototype design and how to optimize airflow and mist distribution for maximum efficiency.”

Student 3 – Tameem Siddiqui Faisal:

“I learned the importance of **photocatalysis and material selection** in air purification. Implementing the TiO₂–CuO honeycomb and UV activation gave me practical exposure to photocatalytic oxidation and its role in pollutant degradation. It also strengthened my skills in experimental setup and performance validation.”

Student 4 – Krish Pradip Kerkar:

“The project helped me develop **problem-solving and teamwork skills**. We faced challenges like water carryover, airflow optimization, and sensor calibration. Overcoming these taught me patience, iterative design, and critical thinking. Overall, the project deepened my understanding of environmental engineering principles and low-cost sustainable solutions.”

10. Visuals:



11. QR Code of Demonstration Video

